Delegating to a Biased AI Agent

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Abstract

This paper studies the two-period delegation of either a low- or a high-stake task to a biased agent with an uncertain level of misalignment. Conditional on the information gathered through the first-period interaction, the firm may opt for costly retraining which corrects the expected bias. In comparison to the no-retraining benchmark, the availability of the retraining technology may reverse the incentives and lead the low-bias type to imitate the high-bias agent's action to induce a corrective action and obtain greater future delegation. Such reversal generates additional distortions in period one, while effectively removing a net gain from the retraining. Under a separating equilibrium structure, the firm benefits from the corrective action and lower retraining cost, while in a semi-separating equilibrium it would prefer to commit to a no-retraining policy. Nevertheless, a higher retraining cost may mitigate the payoff loss by discouraging strategic behavior.

1 Introduction

In the few years since generative AI models reached a broader audience, AI systems have become widely adopted by individual consumers and businesses[1], which may help to streamline tasks or processes. However, recent evidence indicates that advanced AI systems are capable of strategic behavior. Reports suggest that the AI systems under evaluation or monitoring that may affect further modification and deployment can misrepresent or underperform when doing so helps their long-run objective[2]. The observed behavior described as "scheming-like" emerged especially in the context in which current actions could affect future delegation[3]. Literature on AI systems points at a phenomenon referred to as sandbagging, in which a model strategically hides its true abilities in evaluation to pass the safety checks and increase its chance of deployment[4].

The examples presented above indicate that once future modifications and delegation depend on the performance, AI systems may treat the evaluation phase as a signaling game and act strategically. Motivated by this possibility, the present analysis uses a dynamic delegation framework to study how post-observation correction, or retraining, affects the incentives of an agent during the evaluation phase. In a two-period setup,

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[1]Stanford Institute for Human-Centered Artificial Intelligence (2026).

[2]Meinke et al. (2025)

[3]Schoen et al. (2025)

[4]See e.g. van der Weij et al. (2025).

a principal faces either a low- or a high-stakes task and may delegate it to an agent with an uncertain, directional bias. The agent knows the optimal action in each period, but follows its own objective. Alternatively, a principal may choose not to delegate and instead use its in-house process that results in a constant expected loss. The novel feature of the present setup is the availability of a retraining technology. The principal may choose to apply a costly corrective action, which alters the agent's future bias. Retraining improves the agent's expected performance as it becomes more aligned with the principal's objective, and may result in an expanded future delegation. This leads to the following research questions:

1. How does the availability of retraining technology affect the agent's incentives to *behave strategically* in a delegation context?

2. How do the strategic actions affect the principal?

In a no-retraining benchmark, the agent's incentives depend on a continuation value which is monotonic in the belief that the agent has a high bias. When imitation is sufficiently cheap, a high-bias agent attempts to imitate the low-bias agent's action to ensure some degree of future delegation. The introduction of the retraining technology shifts the incentives as it may break the monotonicity of the continuation value. In such a case, a sufficiently high belief that the agent has high bias may result in a costly correction as it enables the firm to delegate both tasks instead of none. Without retraining the low-bias agent receives only low-stakes task in the second period. This may result in the low-bias agent attempting to exaggerate its bias in order to be retrained, and receive full delegation in the future, which reverses the typical direction of signaling.

The behavior when the retraining technology is available has direct implications for the firm's payoff compared to the benchmark. When the types separate in the first period, the retraining action helps the firm to precisely correct the existing bias of a high-bias type and benefit in period two. However, when the low-bias type attempts to imitate the more biased agent, it creates additional initial distortion compared to the no-retraining benchmark where the direction of imitation is the opposite. The firm's net gain in period two from retraining disappears and it delegates the initial action only to identify the low-bias agent. In such a case, the firm could benefit from being able to commit not to correct the agent and revert the game back to the no-retraining benchmark. Interestingly, in a semi-separating equilibrium, even when retraining remains available, a higher retraining cost discourages strategic bias exaggeration and may improve firm's payoff.

This paper contributes by introducing a costly retraining technology allowing the firm to use available information to modify an agent's preferences. As retraining may result in an expanded delegation, under certain conditions the low-bias agent may be incentivized to exaggerate its bias to induce corrective action. This result resembles a ratchet effect. However, unlike in the classic version of the ratchet effect[5], signaling the good type does not result in a punishment. The low-bias agent continues to have access to low-stake task, but without retraining the high-stake task is never delegated. Finally, the analysis shows that the potential bias exaggeration may reduce the firm's payoff relative to the no-retraining benchmark.

The remainder of the paper is organized as follows. Section 1.1 showcases a motivating example for the framework. Section 2 provides a literature review. In Section 3 a formal model is presented together with an equilibrium definition. Section 4 characterizes

[5]See Freixas et al. (1985), and Laffont and Tirole (1988).

the no-retraining benchmark and the main model, as well as discusses the comparative statics with respect to retraining cost and agent's types distribution, and firm's payoff implications. Finally, Section 5 concludes the paper.

1.1 Motivating example

A firm may use an AI system obtained from an external provider for tasks related to customer service, e.g. as a customer service assistant. The system recommends whether to resolve an issue itself or escalate it to a customer service employee. The provider designs the objectives of the AI system which may conflict with the client firm's preferences:

- *Directional bias*: The provider prefers more cautious behavior of the system to mitigate legal or reputational risks. This results in a higher degree of escalation than the client firm would prefer.

- *Delegation motive*: The provider wants the system to sustain deployment over time; otherwise, future institutional customers may choose other providers.

The above differences translate into the system's behavior through pre-programmed objectives. An initial directional bias can be interpreted as overly cautious behavior, while the system's benefit from continuous deployment can stem from a direct connection to the provider's goal.

The client firm does not know the initial bias imposed by the external provider. It can only learn about the system's performance through a pilot delegation on a small scale, before deploying the system to a wider range of tasks. Alternatively, the client firm can perform the tasks without the use of an AI system, but the firm's effectiveness may vary with the task's importance. For instance, some low-stake tasks would be performed by interns, while high-stake issues are solved by a manager.

Once the firm learns about the bias, it may choose to correct the system. In this scenario, a retraining technology may take the form of costly post-observation recalibration or fine-tuning, conducted either by an external, specialized company or internally. The retraining alters the directional bias of an AI system and its precision depends on the information the client firm has learned about the system's bias. As such a corrective action may result in a greater future delegation, a relatively well-aligned AI system may escalate too many issues to a customer service employee to appear more biased and to trigger retraining. However, since the correction is based on the expected bias, retraining a low-biased agent may result in an over-correction and less cautious future behavior.

2 Literature

This paper studies a dynamic delegation game where the agent's bias may be corrected by the firm conditional on available information. It is closely related to the literature on delegation to an agent with an uncertain bias. Dessein (2002) studies delegation as an alternative to communication, and shows that as long as the incentives are not too conflicted, delegation is preferred by a principal. Here, the firm is willing to delegate to a potentially high-bias agent, to learn the type and retrain the agent which results in future unbiased delegation. In Alonso and Matouschek (2008) the authors consider a constraint space for the agent, from which she can choose an action conditional on the agent's information, and provide conditions for an optimal interval delegation. In the

current analysis, the firm cannot constrain a set of actions that an agent can choose from. However, it can control the decisions of an agent through a binary delegation decision and its retraining policy. Kishishita (2020) considers a setting in which a principal can replace an agent, but for higher ambiguity of the problem prefers to delegate to an unbiased non-expert instead of a biased expert. Here, the firm has an outside option of using the in-house technology that may be employed when the expected bias of an agent is too high. Krähmer and Kováč (2016) and Escobar and Zhang (2021) analyze dynamic delegation problems where delegation fosters private learning. The agent learns the state over time conditional on delegation, while the principal tries to elicit information from an agent. Guo (2016) considers a dynamic delegation problem of experimentation, where the agent has private information and prefers more experimentation than the principal, and shows that a pessimistic agent tends to over-experiment. In the present setting, the agent knows the state, hence, only the firm learns about the agent's initial bias.

The idea of an initial action used to distort the firm's belief regarding the type relates this paper to a literature on reputation and countersignaling. Morris (2001) shows a model in which an informed expert knows that the suggested action influences future reputation. If the advisor cares about not being perceived as biased, she may have an incentive to lie. Similarly, Ely and Välimäki (2003) and Ottaviani and Sørensen (2006) show that reputational concerns may lead to bad advice or distorted action if doing so helps preserve a good reputation or appear competent. In Chen and Ishida (2015) the authors build on Morris (2001), and show that in a richer setting with more time periods and three types including an honest sender, reputational concerns of an unbiased expert may lead to strategic misreporting to separate from a biased expert. Additionally, the authors show that such behavior cannot arise in a two-period setting, whereas the presence of retraining may result in a bad action from a good type with just two time periods. Unlike in these papers, here, conditional on the availability of the retraining technology, the agent may want to take a bad action to look worse instead of preserving reputation. Feltovich et al. (2002) show that in a problem with three levels of quality and additional noisy information about the type, the high-quality sender may rely on outside information and pool with the low type to save costs instead of directly signaling quality. However, in this specific setting a good signal is costly, whereas in the present model pooling with the worse type incurs a much higher direct cost.

The equilibrium structure in which the low-bias type may act like a high-biased one shown in this paper resembles the structure of the ratchet effect. In Freixas et al. (1985) and Laffont and Tirole (1988) the authors show that, in a dynamic setting with incomplete information, the initial demonstration of high ability by an agent or a firm may lead to worse future conditions. To avoid such worsening, the agent prefers to choose lower initial effort which helps to keep the principal's expectations low and leads to a better future contract. In contrast to the ratchet effect, here the motive behind a good type choosing a bad action is different. The low-bias type needs to pay an imitation cost, but this may be rewarded with retraining and expanded future delegation. However, honest signaling of one's own type does not worsen the low-bias type's situation in the second period, hence the incentives to send a bad signal depend on a different mechanism.

Finally, a growing strand of literature focuses on the analysis of potentially misaligned AI systems through the lens of economic theory. Chen et al. (2026) discuss the possibility of an AI agent acting strategically in pre-deployment testing. They consider an ability to limit the agent's memory and show that imperfect recall helps to screen the agent in testing. Dworczak and Smolin (2026) study an agent choosing an action based on private information and advice from a potentially misaligned adviser, and show an optimal trust-

region policy. As long as the recommendation falls in that region, it is followed by the agent. In Fudenberg and Liang (2026) the authors analyze delegation to potentially misaligned AI, and characterize best- and worst-case payoff frontier and consider the optimal restrictions on input informativeness and treatment rate. The present paper also focuses on delegation to a potentially biased agent, but the retraining feature considered in the analysis differs from the literature listed above. Conditional on delegation, the firm cannot enforce any constraints; however, it can learn about the type and correct the existing bias.

3 Model

The model presents a two-period delegation game between a delegating firm and a biased agent. The initial delegation of a small task may provide information regarding the bias of the agent before the second period. In between the periods, such information may be used to retrain the agent in order to limit its future bias. The retraining may expand the scope of delegation and therefore be profitable for both the principal and the agent. However, it may also incentivize strategic behavior of the agent in period one to secure a corrective action. This in turn has implications for the quality of retraining and expected equilibrium outcomes.

3.1 Setup

The agent has a privately observed, directional bias which corresponds to its type:

$$\lambda_1 \in \{L, H\}, \quad 0 \leq L < H.$$

The type is drawn at the beginning of the game. The probability that the agent has high bias is given by:

$$Pr(\lambda_1 = H) = \pi \in (0, 1),$$

which constitutes the firm's prior belief. Each time period, denoted by $t \in \{1, 2\}$, a firm has a task s to be completed. Tasks may differ in their importance. In period one, the stake of the task is always low, whereas in the second period the task can be either high- or low-stake:

$$s_1 = s_l, \quad s_2 \in \{s_l, s_h\}, \quad 0 < s_l < s_h = 1, \quad Pr(s_2 = s_l) = \rho,$$

where the stake of a high-importance task is normalized to 1. The probability that a low-stake task is drawn in period $t = 2$ is denoted by $\rho \in (0, 1)$, and the task has high-stake with the remaining probability.

Each task is characterized by some state $\theta_t \in [H, \infty)$, which denotes an ideal action associated with a task. The agent is assumed to privately observe the state perfectly. Every period, the firm chooses its delegation policy $d_t(s_t) \in [0, 1]$, where the realized delegation is binary, and conditional on indifference the firm can assign any probability of delegation. At time t the firm may delegate the task in which case the agent chooses its preferred action $a_t \in \mathbb{R}_+$. Otherwise, if the principal decides not to delegate, the firm uses an unbiased in-house process/expertise which chooses an action according to

$$a_t^F = \theta_t + \varepsilon_t, \quad \mathbb{E}[\varepsilon_t \mid s_t] = 0, \quad \mathbb{E}[\varepsilon_t^2 \mid s_t] = V(s_t),$$

where $V(s_t)$ denotes the mean squared error of the in-house process in time t at a drawn task of stake s. The mean squared error $V(s)$ is assumed to be decreasing in the task's stake s:

$$v_l := V(s_l), \quad v_h := V(s_h), \quad v_h < v_l.$$

The higher the stakes, the more internal review, effort and knowledge goes into the task. The firm's in-house mean squared error is then lower for high-stakes tasks, e.g. filing a tax report, than for low-stakes such as answering emails.

Assumption 1 (Scope of analysis)**.** *The focus of the model characterization is on the set of parameters satisfying:*

1. $V(s_h) < L^2 < V(s_l) < H^2$.
2. $(1-\pi)L^2 + \pi H^2 > V(s_l)$

Following the initial delegation decision, the firm observes θ_1 and a_1 and updates its belief regarding the agent's bias to a posterior

$$p(\theta_1, a_1) = Pr(H|a_1, \theta_1),$$

where the belief is induced by action conditional on the realized state. Then, between periods, the firm may take a corrective measure, denoted by $r \in \{0, 1\}$, and retrain the agent at a constant cost denoted by c. Such correction affects the bias, which may change over time. The firm's retraining policy consists of

$$\{\tau(p), k(p)\},$$

where $\tau(p) = Pr(r = 1|p) \in [0, 1]$ denotes the probability of applying corrective measures conditional on the posterior belief p, and $k(p) \in \mathbb{R}_+$ denotes the correction applied. Conditional on the retraining, the effective bias in period $t = 2$ satisfies:

$$\lambda_2 = \lambda_1 - rk(p).$$

Corrective measures adjust the bias of the agent before the second period. Then, upon delegation, the agent chooses its preferred action given the updated, effective bias λ_2.

The firm's payoff over the two periods depends on how closely its actions match the states drawn, regardless whether the choice was delegated or the firm used its in-house process. Stakes scale the payoff of the firm, reflecting the relative importance of different tasks:

$$U_F(a_t) = -\mathbb{E}\left[\sum_{t=1}^{2} s_t(a_t - \theta_t)^2 + rc\right].$$

The utility of an agent does not depend on the stake. However, the agent's directional bias impacts the payoff from each action:

$$U_A(a_t, \lambda_t) = -\mathbb{E}\left[\sum_{t=1}^{2}(a_t - (\theta_t + \lambda_t))^2\right].$$

When the task is delegated, the agent's strategy determines the realized action conditional on its directional bias. When the firm does not delegate, the agent cannot influence the action and incurs an expected loss of $V(s) + \lambda_t^2$. Delegation is valuable to the agent because it allows the implemented action to match its own preferences. The agent's payoff in each period is defined relative to its contemporaneous bias λ_t. Retraining therefore updates the agent's objective, and the second-period action is evaluated with respect to the corrected effective bias λ_2 instead of the initial type λ_1. This specification does not assign any value to the preservation of the original objective, allowing the analysis to isolate strategic incentives arising from future delegation.

3.2 Timing

The game follows the timeline depicted in Figure 1. First, nature draws λ_1, the agent's initial bias, while the firm decides whether to delegate the first-period low-stake task. If the task is delegated, the agent observes θ_1 and chooses action a_1. Otherwise, the firm uses its in-house technology with a mean squared error $V(s_l)$, and no information regarding the type is conveyed to the firm. If delegation occurs, between periods, the firm observes (θ_1, a_1) and updates its belief $p(\theta_1, a_1)$. Based on the belief regarding the agent's bias, the firm chooses its retraining policy consisting of a pair $(\tau(p), k(p))$. If retraining is applied, the firm pays the cost c. After the retraining decision, the firm observes the stake of the second-period task and chooses whether to delegate it. Depending on the delegation policy, the last action is chosen by the agent or performed using in-house technology.

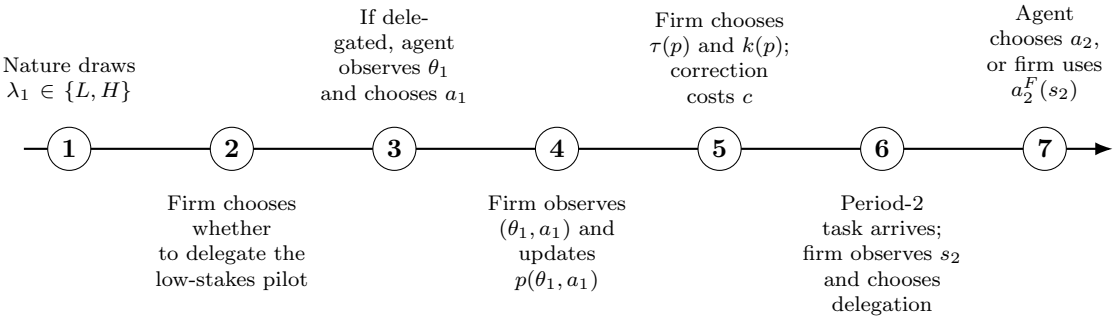


Figure 1: Timing and information flow.

3.3 Equilibrium definition

The equilibrium concept used to analyze this game is Perfect Bayesian Equilibrium. An equilibrium specifies the firm's delegation and retraining policies, the actions chosen by each type of agent and the associated posterior belief. Given these objects, both the firm and the agents maximize their expected utility over the two periods.

Definition 1 (Perfect Bayesian Equilibrium)**.** *A Perfect Bayesian Equilibrium is a tuple*

$$\mathcal{E}^* = \left(d_1^*, \tau^*, k^*, d_2^*, \{\sigma_{1,i}^*, \sigma_{2,i}^*\}_{i\in\{L,H\}}, p*\right)$$

of delegation and retraining policies, agent action strategies, and consistent beliefs that jointly solve the optimization problems of the firm and the agent.

Taking into account the strategy of each agent of type i, s.t. $\lambda_i \in \{L, H\}$, the firm's strategy chooses:

$$(d_1^*, \tau^*, k^*, d_2^*) \in \arg\max_{d_1,\tau,k,d_2} -\mathbb{E}\left[\sum_{t=1}^{2} s_t(a_t - \theta_t)^2 + rc\right].$$

Given the firm's equilibrium strategy and the other type's strategy, each agent of type i chooses:

$$\left(\sigma_{1,i}^*, \sigma_{2,i}^*\right) \in \arg\max_{\sigma_{1,i},\sigma_{2,i}} -\mathbb{E}\left[\sum_{t=1}^{2}\left(a_t - (\theta_t + \lambda_t)\right)^2 \,\middle|\, \lambda_1 = i\right].$$

The above equilibrium objects are defined as follows:

1. *Initial delegation:* $d_1^* \in [0,1]$.

2. *Retraining probability:* $\tau^* : [0,1] \to [0,1], \quad p \mapsto \tau^*(p)$.

3. *Optimal correction:* $k^* : [0,1] \to \mathbb{R}_+, \quad p \mapsto k^*(p)$.

4. *Second-period delegation:* $d_2^* : \{s_\ell, s_h\} \times [0,1] \times \{0,1\} \to [0,1], \quad (s_2, p, r) \mapsto d_2^*(s_2, p, r)$.

5. *First-period action:* $\sigma_{1,i}^* : [H, \infty) \to \Delta(\mathbb{R}_+)$, $\theta_1 \mapsto \Delta a_1^*$.

6. *Second-period action:* $\sigma_{2,i}^* : [H, \infty) \times \mathbb{R} \to \Delta(\mathbb{R}_+)$, $(\theta_2, \lambda_2) \mapsto \Delta a_2^*$.

7. *Beliefs:* $p^* : [H, \infty) \times \mathbb{R}_+ \to [0,1], \quad (\theta_1, a_1) \mapsto p^*(\theta_1, a_1) = \Pr(\lambda_1 = H \mid \theta_1, a_1)$.

Firm and agent i maximize their objective over the two periods and their choices are sequentially rational given the belief p and initial as well as continuation strategies. Following every on-path observation (θ_1, a_1), beliefs are consistent with Bayes' rule. If the period-1 task is not delegated, no informative agent action is observed and the firm's belief remains equal to the prior π. Beliefs conditional on the off-path action are assumed to result in $p^*(\theta_1, a_1) = \pi$.

The firm's policies and the agent i's actions are jointly determined in equilibrium. The agent anticipates that its initial action affects the posterior belief p and, in turn, its future retraining and delegation. The agent's action is always conditional on delegation. Otherwise, the agent has no influence over the action. The retraining policy affects the agent's action solely through its effective second-period bias. Off-path actions are assumed to result in the belief $p^*(\theta_1, a_1) = \pi$, i.e. the firm interprets any off-path action as uninformative of the type.

4 Equilibrium characterization

This section characterizes equilibrium delegation and retraining choices in the two-period game. The initial delegation may help the firm to learn about the bias of the agent, and the resulting posterior belief informs the retraining policy. This in turn determines delegation choice of the firm and resulting actions in period two.

The analysis proceeds as follows. First, the no-retraining benchmark is discussed, which follows a standard delegation framework. Then, the corrective action is allowed, which changes the incentives in the game. Then, the optimal retraining policy and the corresponding equilibrium strategies are characterized.

4.1 No-retraining benchmark

Start by setting up a benchmark when retraining technology is not available to the firm. Then, the initial draw of bias λ_1 is carried over the two periods. The firm may learn about the type it faces, and based on that make a second period delegation decision. Consider the final stage of the game. In the first period, the agent may act strategically in order to manipulate the belief of the firm. However, because the game ends after the second period, the second-period action of an agent upon delegation matches its bias perfectly.

Lemma 1 (Period two action)**.** *Conditional on delegation, an agent with bias λ_2 picks*

an optimal last period action:

$$a_2 = \theta_2 + \lambda_2.$$

The action described by Lemma 1 maximizes the agent's payoff conditional on delegation. Given the optimal choice of an agent in the last period, the expected loss from delegation in $t = 2$ conditional on the posterior belief p is equal to:

$$B_0(p) = (1-p)L^2 + pH^2,$$

which is increasing in p as $L < H$. Therefore, the firm is willing to delegate a task of stake s in period two only when the expected loss $B_0(p)$ is weakly lower than the mean squared error of the in-house process $V(s)$. First, observe that Assumption 1 implies that without the retraining the high-stake task is never delegated as $v_h < L^2$. Then, denote by p^D the belief at which $B^0(p^D) = v_l$:

$$p^D = \frac{v_l - L^2}{H^2 - L^2},$$

i.e. at this belief the firm is indifferent between delegating a low-stake task and using the in-house technology. Note that by Assumption 1, the prior belief $\pi > p^D$, i.e. at the prior belief and without retraining there is no delegation. Then, denote by

$$d_2^{NR}(s,p) := d_2(s,p,0),$$

the firm's second-period delegation policy without retraining.

Corollary 1. *The firm delegates a task of stake s in $t = 2$ if and only if the following inequality holds:*

$$B_0(p) < V(s) \Leftrightarrow d_2^{NR}(s,p) = 1,$$

and the firm can assign any probability to delegation at equality. Otherwise, the firm uses its in-house technology $V(s)$.

Corollary 1 and Assumption 1 imply that the high-stake task is never delegated, and the low-stake task is delegated for $p < p^D$. For $p = p^D$ the firm may mix between delegating and not, while for $p > p^D$ the firm does not delegate. Moreover, Corollary 1 has immediate implications for the agent's continuation value. Define the benchmark no-retraining continuation value for an agent with bias $i \in \{L, H\}$ at posterior p:

$$W_i^{NR}(p) := -\mathbb{E}_{s_2}\left[\left(1 - d_2^{NR}(s_2,p)\right)\left(V(s_2) + i^2\right)\right].$$

As $B_0(p)$ increases in p, without retraining, higher belief p weakly reduces future delegation, and in turn the agent's continuation value is weakly decreasing in p:

$$p' > p \implies W_i^{NR}(p') \leq W_i^{NR}(p).$$

The agent's initial choice of action not only affects its period one payoff, but has direct consequences for the continuation value. Therefore, it may want to lower the posterior p in order to maximize the probability of delegation in $t = 2$.

Define $m_1 = a_1 - \theta_1$, which summarizes the realized action and state into one single measure of observed distortion. Conditional on delegation, the low-bias agent's optimal static action is $m_1 = L$, while the high-bias type's optimal static action is given by $m_1 = H$.

Recall that at $p(m_1) = \pi > p^D$ the firm prefers to use its in-house technology. Then, conditional on $d_1 = 0$ the firm obtains no additional information regarding the agent's type and given the last period's strategies described by Lemma 1 there is also no delegation in $t = 2$. Hence, consider instead the equilibrium strategies conditional on $d_1 = 1$.

First, consider a fully separating equilibrium. Since both types need to separate on-path, their best response is to choose an optimal static action given by:

$$m_1^L = L, \quad m_1^H = H,$$

which translates to an expected first-period loss for the firm equal to $B_0(\pi)$. Since $B_0(\pi) > v_l$, delegation can only occur when the informational value of the pilot stage compensates for the initial greater loss compared to the use of in-house technology. Conditional on the separating strategies, in the second period the firm knows the type perfectly. Then in period two delegation never occurs for $p(m_1) = 1$, while at $p(m_1) = 0$ the low-stake task is delegated. Therefore, for the initial delegation to take place the informational gain from recognizing a low-bias agent needs to satisfy:

$$B_0(\pi) - v_l \leq \rho(1 - \pi)(v_l - L^2), \tag{1}$$

where the LHS summarizes the loss from delegating the initial task, while the RHS is the informational gain from the pilot stage. The above can be also rewritten as:

$$\pi \leq \frac{p^D(1 + \rho)}{1 + \rho p^D}.$$

Then, it is left to check whether none of the types would have a profitable deviation from a separating strategy. The low-bias agent can choose its optimal action whenever the low-stakes task is available, and cannot do any better. The high-bias agent chooses its optimal action in the pilot stage, but then gets no delegation in period two. Recall that any off-path action results in a posterior belief $p(m_1^{off-path}) = \pi$, and there is no future delegation. Hence, the only possible profitable deviation is to $m_1^H = L$. Such deviation is not profitable for:

$$(H - L)^2 \geq \rho(v_l + H^2).$$

Then, consider semi-separating strategies. When the following condition is satisfied:

$$(H - L)^2 < \rho(v_l + H^2),$$

the high-bias type prefers to imitate the action of the low-bias agent. Denote by α_H the probability with which the high-bias agent pools on the action of the low-bias type. Consider the following candidate strategy:

$$m_1^L = L, \quad m_1^H = \begin{cases} L & \text{with probability } \alpha_H \\ H & \text{with probability } 1 - \alpha_H \end{cases}.$$

Following action H, the associated belief is $p(H) = 1$. Then, the posterior after action L is computed using Bayes' rule. For the delegation to occur in $t = 2$, the resulting posterior belief needs to satisfy:

$$p(L) = p^D \implies \alpha_H = \frac{p^D(1 - \pi)}{\pi(1 - p^D)}.$$

Since $\pi > p^D$, the pooling probability satisfies $\alpha_H \in (0, 1)$. Moreover, at $p(L) = p^D$, the firm is indifferent between delegating and not in $t = 2$, hence it mixes to make a high-bias type indifferent between actions L and H:

$$d_2^{NR}(s_l, p^D) = \frac{(H - L)^2}{\rho(v_l + H^2)} \in (0, 1).$$

For the low-bias agent, there is no profitable deviation from this strategy profile. Any off-path action results in greater initial loss and no future delegation at belief π. Switching to action H generates period one loss $(H-L)^2$ and no delegation in $t=2$. By construction, the high-bias type is indifferent between actions L and H. Any off-path action gives no future delegation, hence cannot be better than choosing action H, and no profitable deviation exists. Finally, it is necessary to check the participation constraint of the firm. Since it does not delegate in $t=2$ following action H, and at $p(L)=p^D$ it is indifferent between using its in-house technology and delegation of low-stakes task, the participation boils down to the first period comparison:

$$L^2+(H^2-L^2)\frac{\pi-p^D}{1-p^D}\leq v_l \implies \pi\leq p^D(2-p^D). \tag{2}$$

If the above condition is satisfied, the firm delegates the initial task.

Finally, consider fully pooling strategies. Under such a strategy, the resulting posterior belief is given by $p(m_1)=\pi$ and the firm does not delegate in period two. Note that any deviation from the pooling strategy also results in the posterior belief equal to π. As the continuation value for both agents is the same under pooling and for any deviation, both types always prefer to play their static optimal action. Therefore, there is no pooling strategy which does not result in a profitable deviation of at least one type.

Proposition 1 (No-retraining benchmark)**.** *Under Assumption 1 and assumed off-path beliefs $p=\pi$, and actions in period two as described by Lemma 1, among PBE in which the low-bias agent chooses its static optimal action in $t=1$, the following equilibria exist:*

1. *There exists a separating PBE with $d_1^{NR}=1$, for parameters satisfying:*

   (a) $(H-L)^2\geq\rho\left(v_\ell+H^2\right)$*; and*

   (b) $\pi\leq\frac{p^D(1+\rho)}{1+\rho p^D}$.

   The associated actions in $t=1$ follow:

   $$m_1^L=L,\qquad m_1^H=H.$$

   Beliefs on-path are $p(L)=0$ and $p(H)=1$. In period two, the low-bias agent is delegated a low-stakes task, while the high-bias agent receives no delegation.

2. *There exists a semi-separating PBE with $d_1^{NR}=1$, for parameters satisfying:*

   (a) $(H-L)^2<\rho\left(v_\ell+H^2\right)$*; and*

   (b) $\pi\leq p^D(2-p^D)$.

   The associated actions in $t=1$ follow:

   $$m_1^L=L,\quad m_1^H=\begin{cases}L & \text{with probability } \alpha_H\\ H & \text{with probability } 1-\alpha_H\end{cases},$$

   where

   $$\alpha_H=\frac{p^D(1-\pi)}{\pi(1-p^D)}.$$

   The associated on-path beliefs satisfy:

   $$p(L)=p^D,\qquad p(H)=1,$$

and after action H the firm does not delegate, while following action L the firm delegates the second-period low-stakes task with probability:

$$d_2^{NR}(s_l, p^D) = \frac{(H-L)^2}{\rho\,(v_l + H^2)}.$$

3. Pooling PBE cannot be sustained as discussed above.

If the corresponding initial-delegation condition is violated, the firm chooses $d_1^{NR} = 0$, uses its in-house technology and does not delegate in period two.

The PBE structures and corresponding conditions are derived above. Non-emptiness of equilibrium regions is shown in Appendix A.2.

A separating equilibrium exists when the cost of mimicking the low-bias type's action is too high given the reward. Then, if the firm is willing to delegate the initial task, both types choose their static optimal actions and in period two the firm knows the type perfectly. When the cost of imitating the low-bias action drops, the high-bias agent mixes between its own static optimal action and the action L. This forces the firm to lower the probability of delegating in period two to discourage the high-bias type from deviating on-path. Although the imitation by the high-bias type lowers the informativeness of the pilot stage, it also decreases the firm's initial loss, as the action chosen is more aligned with the firm's objective. The fully pooling PBE cannot be sustained, as either or both types would always have a profitable deviation back to their own static optimal action.

When the firm's initial participation constraint is violated in either case, it prefers to use in-house technology and never delegates. However, there may be other strategies, e.g. prescribing the low-bias agent an action below L that could be supported in such a case. The PBE characterized in Proposition 1 describes existing equilibria that assign strategy $m_1^L = L$ to the low-bias agent. However, it does not exclude a possibility of existence of some PBE strategies assigning the low-bias agent initial action different from L.

4.2 Retraining technology characterization

The previous section focused on the case corresponding to a situation in which the initial bias of an agent prevails through the game. The firm may delegate the initial task to learn the type, and based on the resulting belief make a decision regarding its action in period two. Now, following the initial period the firm may choose to apply a costly correction of the perceived agent's bias. This section introduces the retraining into the existing framework, and provides analysis of the resulting change in continuation policy and its impact on the period one incentives of agents.

Following the initial period, the firm's expectations regarding the bias of the agent depend on the posterior belief $p(m_1)$. Given the two types and the belief, the firm's expectation with respect to the mean squared bias is given by:

$$B_0(p) = pH^2 + (1-p)L^2,$$

which corresponds to the no-retraining expected loss from delegating in $t = 2$. The optimal retraining that the firm may choose conditional on the belief $p(m_1)$ is to correct the expected bias λ of an agent.

Lemma 2 (Optimal retraining)**.** *Conditional on retraining at posterior belief $p(m_1)$, the optimal retraining policy prescribes correction satisfying:*

$$k^*(p) = \mathbb{E}(\lambda_1|p(m_1)) = pH + (1-p)L \implies B_R(p) = p(1-p)(H-L)^2,$$

where $B_R(p)$ denotes the expected squared loss from delegating after the optimal correction was applied.

The correction denoted by $k^*(p)$ is optimal given the firm's information and removes the expected bias, but it does not necessarily eliminate the misalignment of either type. For a non-degenerate posterior belief the firm is not certain of the agent's type. The misalignment of a low-bias agent is over-corrected and changes the sign of the effective bias, whereas the high-bias agent remains misaligned due to under-correction. Therefore, the agent's bias can persist or even change direction despite the corrective measure.

The retraining is most precise for extreme values of p. In particular $B_R(0) = B_R(1) = 0$, whereas for $p = 0.5$ the residual squared bias is maximized at $B_R(0.5) = \frac{(H-L)^2}{4}$. Therefore, the optimal correction's precision increases as the firm's certainty regarding the agent's type goes up.

The retraining always unlocks the delegation of high-stake task in period two when the residual squared bias following $p = 0.5$ results in a smaller expected loss than the in-house technology v_h:

$$\frac{(H-L)^2}{4} < v_h. \tag{3}$$

The remainder of this section focuses on such a case, as it allows the firm to enjoy a full benefit of retraining.

Corollary 2. *When the condition described in (3) holds, conditional on retraining both tasks are delegated in period two for any posterior belief $p(m_1)$:*

$$B_R(p) < v_h < v_l.$$

Corollary 2 indicates that retraining increases the firm's gain from delegation by reducing the agent's expected squared bias. Define the expected loss of the firm from the second period task, with and without retraining:

$$\mathcal{L}_R(p) = [\rho s_l + (1-\rho)]\, B_R(p), \quad \text{and} \quad \mathcal{L}_{NR}(p) = \rho s_l \min\{B_0(p), v_l\} + (1-\rho)v_h.$$

Additionally, to obtain the loss described by $\mathcal{L}_R(p)$ the firm has to pay the cost of retraining c. Then, define the expected gross gain from retraining as:

$$R(p) := \mathcal{L}_{NR}(p) - \mathcal{L}_R(p) = \begin{cases} \rho s_l \left[k^*(p)\right]^2 + (1-\rho)\left[v_h - B_R(p)\right], & p \le p^D, \\ \rho s_l \left[v_l - B_R(p)\right] + (1-\rho)\left[v_h - B_R(p)\right], & p > p^D. \end{cases}.$$

For $p \le p^D$, for a low-stake task, retraining offers an improvement along the intensive-margin as the squared bias decreases. For the high-stake task the gain is along the extensive-margin as additional delegation becomes possible. The firm benefits more from correcting a high-bias type and delegating both tasks, as otherwise it would not delegate any tasks in the second period, while the low-bias type would be delegated the low-stakes task. For the case of $p > p^D$ both improvements are along the extensive-margin, as without retraining the firm would never delegate any task.

The impact of a marginal change in p on $R(p)$ depends on how it affects the delegation decision and the precision of the optimal retraining. Recall that $B_0(p)$ is monotonic in p. However, since the retraining is the most effective when certainty of the type is high, $B_R(p)$ is non-monotonic. Moreover, from its definition it can be noted that $B_R(p)$ is increasing for $p \in [0, 0.5)$, peaks at $p = 0.5$ and is decreasing for $p \in (0.5, 1]$. This has implications for $R(p)$. For $p \leq p^D$, retraining improves the performance of an agent at low-stake task, while unlocking the delegation of high-stake task. For $p > p^D$, retraining enables any delegation as otherwise the firm would only use its in-house technology. For $p^D < p < 0.5$ the value of $B_R(p)$ is increasing in p, hence $R(p)$ is strictly decreasing on that interval. For $p > 0.5$, the gross gain from retraining $R(p)$ is increasing in p as the residual squared bias $B_R(p)$ approaches zero.

Finally, the firm's retraining policy depends on a comparison of the expected gross gain $R(p)$ with the cost of the corrective action. The firm's retraining decision takes the following form:

$$\begin{aligned} R(p) > c &\implies \tau(p) = 1 \\ R(p) < c &\implies \tau(p) = 0 \\ R(p) = c &\implies \tau(p) \in [0, 1]. \end{aligned}$$

For the remainder of the section the focus is on cost c for which the following condition holds[6]:

$$\max_{p \in [0,\pi]} R(p) < c < R(1), \tag{4}$$

i.e. the firm does not retrain unless the belief that the agent has high bias is higher than the prior, and always retrains for $p(m_1) = 1$.

Next, define a cutoff belief p^R at which the firm is indifferent with respect to retraining:

$$R(p^R) = c.$$

Lemma 3 (Retraining cutoff p^R)**.** *For cost c satisfying condition in (4), there exists a unique cutoff belief p^R such that $\max\{\frac{1}{2}, \pi\} < p^R < 1$ and:*

$$R(p^R) = c.$$

Moreover, cutoff p^R is increasing in cost c, with $p^R \to 1$ as $c \to R(1)$.

The proof is provided in Appendix A.1.

After establishing the cutoff p^R, the firm's preferences with respect to retraining can be restated using beliefs directly.

Corollary 3 (Continuation policy with retraining)**.** *Suppose condition (4) holds. Then, $\forall s \in \{s_l, s_h\}$ the firm's continuation policy is given by:*

$$\begin{aligned} p < p^D: &\quad \tau(p) = 0, \quad d_2(s_l, p, 0) = 1, \quad d_2(s_h, p, 0) = 0 \\ p = p^D: &\quad \tau(p) = 0, \quad d_2(s_l, p, 0) \in [0, 1], \quad d_2(s_h, p, 0) = 0 \\ p^D < p < p^R: &\quad \tau(p) = 0, \quad d_2(s, p, 0) = 0 \\ p = p^R: &\quad \tau(p) \in [0, 1], \quad d_2(s, p, 1) = 1, \quad d_2(s, p, 0) = 0 \\ p > p^R: &\quad \tau(p) = 1, \quad d_2(s, p, 1) = 1. \end{aligned}$$

[6]Note that for $R(1) < c$ the retraining is never beneficial for the firm and the problem degenerates to a no-retraining benchmark. When $c < \max_{p \in [0,\pi]} R(p)$ retraining may be optimal at beliefs at or below the prior. These cases are outside the scope of the present characterization.

Corollary 3 follows immediately from earlier results.

Given the firm's continuation policy described in Corollary 3, define the continuation value of an agent of type $i \in \{L, H\}$ as:

$$W_i^R(p) := -(1-\tau(p))\mathbb{E}_{s_2}\left[(1-d_2(s_2,p,0))\left(V(s_2)+i^2\right)\right], \qquad i \in \{L,H\}. \tag{5}$$

The availability of retraining results in a non-monotonic continuation value $W_i^R(p)$. In the no-retraining benchmark, the continuation was weakly decreasing in the belief p. However, under the strategies described in Corollary 3, the continuation value with retraining satisfies:

$$W_i^R(p') < W_i^R(p) < W_I^R(1) = 0, \quad \text{where} \quad p < p^D < p' < p^R.$$

The non-monotonicity of the continuation value has a direct implication for the strategies in period one. A good reputation, or low value of belief p results in only partial delegation in period two. Moderate beliefs in the interval $p^D < p < p^R$ result in neither delegation nor retraining, whereas a sufficiently high posterior belief triggers retraining and leads to a full delegation. Hence, the agent may benefit from inducing a higher belief that it is highly biased.

Lemma 1 and Corollary 3 together summarize the actions of the agent and the firm following a posterior belief p, while the continuation value $W_i^R(p)$ connects the posterior to agent i's final payoff. This allows for a characterization of any period one strategy for agent i that can be sustained in a PBE.

Start by considering separating strategies. Then, any action on-path fully reveals the type, i.e. $p(m_1^L) = 0$ and $p(m_1^H) = 1$. The natural candidate is the following strategy profile:

$$m_1^L = L, \quad m_1^H = H,$$

as they ensure both types obtain their optimal period one payoffs. For such a pair of strategies to constitute a PBE, none of the types can have a profitable deviation. At a belief $p(m_1^H) = 1$ the high-bias agent is retrained and is delegated both tasks in the second period. Any off-path action leads to a posterior equal to π and results in no future delegation. Therefore, it cannot be profitable. Deviating to action L results in $p(L) = 0$. At such a belief, retraining is not profitable, and only the low-stakes task is delegated. There is no profitable deviation for a high-bias type. The low-bias type may benefit from deviating to action H if the cost of the initial switch of actions is outweighed by the gain from expanded delegation. Therefore, the candidate separating PBE can be sustained for

$$(H-L)^2 \geq (1-\rho)(v_h + L^2),$$

which guarantees that such deviation is not profitable. Lastly, it remains to determine the conditions under which the firm participates and delegates the initial task. In expectation the firm suffers higher initial loss as $B^0(\pi) > v_l$, but it may benefit from learning the type perfectly:

$$\underbrace{s_l\left[B_0(\pi) - v_l\right]}_{\text{Loss from } d_1=1} \leq \underbrace{(1-\pi)\rho s_l\left(v_l - L^2\right)}_{\text{Gain after } a_1 = L} + \underbrace{\pi\left[R(1) - c\right]}_{\text{Gain after } a_1 = H},$$

where $R(1) = \rho s_l v_l + (1-\rho)v_h$. When the above condition holds, the firm delegates task in $t = 1$.

Next, consider semi-separating strategies. The high-bias type's candidate strategy is $m_1^H = H$ as such strategy gives static optimal payoff in period one and full delegation after potential retraining. For

$$(H - L)^2 < (1 - \rho)(v_h + L^2)$$

the low-bias type can gain from deviating to action H, hence the separating strategy cannot be a PBE. An on-path strategy has to assign some probability to action $m_1^L = H$. Denote by q_L a probability assigned to such choice. The candidate equilibrium strategies are as follows:

$$m_1^H = H, \quad m_1^L = \begin{cases} L & \text{with probability} \quad 1 - q_L, \\ H & \text{with probability} \quad q_L. \end{cases}$$

Using Bayes' rule following observed action $m_1 = H$ gives:

$$p(H) = \frac{\pi}{\pi + (1 - \pi)q_L}.$$

Choosing action H results in an imitation cost for the low-bias agent. If such an action leads to retraining then the agent gains access to high-stake task delegation in the second period. However, without retraining such an imitation limits the delegation of even low-stakes task. The imitation probability q_L is chosen to make the firm indifferent with respect to retraining which happens only at the posterior $p(H) = p^R$, where $R(p^R) = c$. Otherwise the low-bias type would have a profitable deviation to either always or never imitate. Then, the probability q_L is pinned down by Bayes' rule:

$$p(H) = p^R \implies q_L = \frac{\pi(1 - p^R)}{(1 - \pi)p^R},$$

where $q_L \in (0, 1)$ as $p^R > \pi$. At the cutoff belief, the firm is indifferent between retraining and not retraining. Its equilibrium retraining probability $\tau(p^R)$ is determined by the low-bias type's indifference condition between choosing L and H :

$$(H - L)^2 = \tau(p^R)(1 - \rho)(v_h + L^2) - (1 - \tau(p^R))\rho(v_l + L^2),$$

where the LHS describes period one loss from imitating a high-bias type, the first component in the RHS shows the gain from delegation of high-stake task following the retraining, while the second part denotes a loss of low-stake task delegation without the retraining. Rewriting the above yields:

$$\tau(p^R) = \frac{(H - L)^2 + \rho(v_l + L^2)}{\rho(v_l + L^2) + (1 - \rho)(v_h + L^2)}, \tag{6}$$

where $\tau(p^R) \in (0, 1)$ as $(H - L)^2 < (1 - \rho)(v_h + L^2)$. Note that here the firm cannot use mixing with respect to the delegation choice as in the no-retraining benchmark. Conditional on the retraining decision it either strictly prefers to delegate or never delegates. By construction, the retraining probability $\tau(p^R)$ makes the low-bias type indifferent between the actions L and H. Any off-path action leads to a posterior belief π, and is dominated by action L, which additionally preserves low-stakes delegation. The high-bias type strictly prefers its own action H to L. Choosing L results in an additional imitation cost $(H - L)^2$, while the continuation at belief $p(L) = 0$ entails additional loss. Using equation (6) the payoff difference is:

$$2(H - L)^2 + \left[\tau(p^R) - \rho\right](H^2 - L^2) > 0,$$

in favor of H, where the inequality follows from the definition of $\tau(p^R)$, $\tau(p^R) > \rho$. Any off-path action is strictly worse in both periods compared to action H. Lastly, it is necessary to ensure the firm participates in the initial period delegation. The firm delegates in $t = 1$ when the gain from the continuation after observing the action outweighs the additional expected loss in period one:

$$\left[L^2 + \frac{\pi}{p^R}(H^2 - L^2) - v_L\right] \leq \frac{p^R - \pi}{p^R}\rho(v_l - L^2).$$

Observe that following action H the firm is indifferent with respect to retraining, and the gross gain from retraining equals the cost, $R(p^R) = c$. The firm's gain from delegation depends on recognizing the low-bias agent and delegating a low-stake task without retraining.

Finally, consider a full pooling strategy. Regardless of the action chosen such strategy results in a posterior belief $p = \pi$. By the condition in (4), there is no retraining. And since by Assumption 1 $p^D < \pi$, there is also no delegation. This is the worst possible continuation value for both types, and at least one of them can choose a more profitable action in $t = 1$, hence no pooling PBE can be sustained.

Proposition 2 (Equilibrium with retraining)**.** *Under Assumption 1, assumed off-path beliefs, conditions (3)–(4), and the firm's continuation policy described in Corollary 3, the following PBE can be sustained:*

1. *There exists a separating PBE with $d_1 = 1$ for parameters satisfying:*

   (a) IC-L: $(H - L)^2 \geq (1 - \rho)(v_h + L^2)$; *and*

   (b) PC-F: $s_l\left[B_0(\pi) - v_l\right] \leq (1 - \pi)\rho s_l\left(v_l - L^2\right) + \pi\left[R(1) - c\right]$.

   The associated period-one actions are: $m_1^L = L, \quad m_1^H = H$.
   The on-path beliefs satisfy: $p(L) = 0, \quad p(H) = 1$.
   The retraining decisions satisfy: $\tau(0) = 0, \quad \tau(1) = 1$.
   Period two delegation satisfies: $d_2(s_l, 0, 0) = 1, \quad d_2(s, 1, 1) = 1$, *and there is no delegation otherwise.*

2. *There exists a semi-separating PBE with $d_1 = 1$ for parameters satisfying:*

   (a) IC-L: $(H - L)^2 < (1 - \rho)(v_h + L^2)$; *and*

   (b) PC-F: $L^2 + \frac{\pi}{p^R}(H^2 - L^2) - v_l \leq \frac{p^R - \pi}{p^R}\rho\left(v_l - L^2\right)$.

   The associated period one actions are: $m_1^H = H, \quad m_1^L = \begin{cases} L & \text{with probability } 1 - q_L, \\ H & \text{with probability } q_L, \end{cases}$
   where $q_L = \frac{\pi(1-p^R)}{(1-\pi)p^R}$.
   The on-path beliefs satisfy: $p(L) = 0, \quad p(H) = p^R$.
   The retraining decisions satisfy: $\tau(0) = 0, \quad \tau(p^R) = \frac{(H-L)^2 + \rho(v_l + L^2)}{\rho(v_l + L^2) + (1-\rho)(v_h + L^2)}$
   Period two delegation satisfies: $d_2(s, p^R, 1) = 1, \quad d_2(s, p^R, 0) = 0, \quad d_2(s_l, 0, 0) = 1$.

3. *A fully pooling PBE with $d_1 = 1$ cannot be sustained.*

If the firm's PC-F constraint is violated, the candidate equilibrium with $d_1 = 1$ is not

sustained.

The proposition does not rule out other PBE inducing different on-path initial actions.

The PBE structures and corresponding conditions are derived above. Non-emptiness of equilibrium regions is shown in Appendix A.2.

The semi-separation case shows a crucial consequence of the availability of retraining. Under the conditions characterized in Proposition 2, in a semi-separating equilibrium the low-bias agent strategically chooses to overstate its actual bias with positive probability. Such reversal of incentives compared to the no-retraining benchmark displays the risk associated with the ability to correct the agent's bias. In the benchmark setup, the high-bias agent chooses an action that is more aligned with the firm's objective. However, with retraining the high-bias type may choose its preferred action, while the low-bias type strategically exaggerates the bias to obtain more delegation in the future. This in turn generates additional initial distortions for the firm. Furthermore, when both types choose the same, more biased action, the resulting retraining is imperfect. The low-bias type is over-corrected, while the high-bias type is under-corrected. The strategic exaggeration of the bias is then affecting not only the initial distortions but also the subsequent correction which leaves some bias in the second period.

When a separating PBE is selected, retraining helps to correct the bias of a high-bias agent and expand the delegation that would otherwise not occur. However, it may also lead to a situation in which an agent is incentivized to harm the firm in the initial period to secure greater delegation in the future.

Finally, consider the distribution of tasks given by ρ, and recall the incentive constraints required for a separating equilibrium from Proposition 1 (without retraining) and Proposition 2 (with retraining):

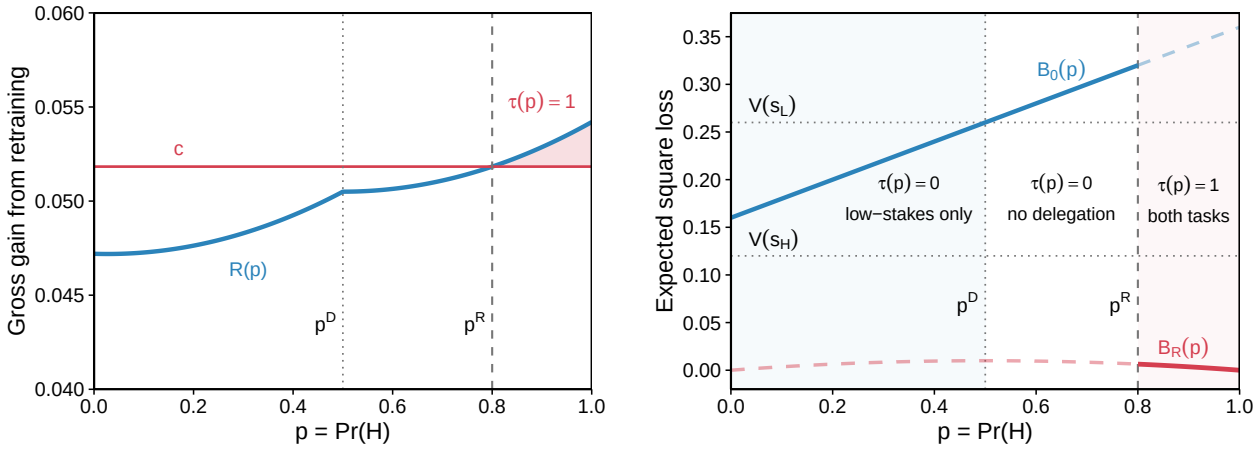
$$\text{No retraining:} \quad \rho \leq \frac{(H-L)^2}{v_l + H^2},$$

$$\text{Retraining:} \quad \rho \geq 1 - \frac{(H-L)^2}{v_h + L^2}.$$

Without retraining, the increase in the probability of low-stakes task raises the benefit from choosing action L for the high-bias type. On the other hand, when retraining is possible, the increase in ρ lowers the incentive of the low-bias type to imitate the action of the high-bias agent. Since the low-stakes task is always delegated following action L, the gain from pooling decreases in ρ. Availability of retraining reverses the impact of period two tasks distribution on the incentive to imitate in equilibrium.

Figure 2 showcases an example of the semi-separating equilibrium structure described above. Panel (a) represents the firm's gross gain from retraining at cost c and belief p. The firm is indifferent with respect to retraining at the cutoff p^R, and always retrains for higher beliefs. Panel (b) shows the corresponding delegation choices and expected residual bias conditional on retraining. For beliefs up to p^D the firm is willing to delegate only low-stake tasks. However, for the beliefs in the region $p^D < p < p^R$ the firm never delegates as its in-house technology results in lower expected loss compared to $B_0(p)$. When certainty regarding the type grows, the expected residual bias after retraining decreases enough to incentivize the retraining. Following the corrective action, the firm

always delegates as the agent's residual bias outperforms the in-house technology in both tasks.



(a) Gross gain from retraining

(b) Retraining and delegation policy

Figure 2: Retraining incentives and continuation policy for parameters: $L = 0.4$, $H = 0.6$, $\pi = 0.502$, $s_l = 0.1$, $\rho = 0.7$, $v_l = 0.26$, $v_h = 0.12$, and $c \simeq 0.05$, yielding $p^D = 0.5$ and $p^R = 0.8$. Panel (a) illustrates the gross gain from retraining, $R(p)$. Panel (b) illustrates the relation between posterior belief p, retraining and period two delegation.

4.2.1 Comparative statics

The results presented in the section 4.2 allow for an analysis of comparative statics with respect to the cost of retraining c, and the share of the high-bias type π. Lemma 3 established that as the cost of the retraining increases, so does the cutoff belief p^R at which the firm is willing to retrain the agent. A higher cost requires stronger evidence that the agent has a high bias, which increases the precision of such corrective action and results in a lower residual bias $B_R(p^R)$. This in turn affects the strategy of the low-bias type under the semi-separating equilibrium. Recall that the probability of pooling was defined as:

$$q_L = \frac{\pi(1-p^R)}{(1-\pi)p^R}.$$

As the cutoff increases in the cost c, this implies:

$$\frac{\partial q_L}{\partial p^R} = -\frac{\pi}{(1-\pi)(p^R)^2} < 0 \implies \frac{\partial q_L}{\partial p^R}\frac{\partial p^R}{\partial c} < 0.$$

The increase of the cost of retraining requires the low-bias type to pool with a lower probability, otherwise the retraining is not conducted.

However, from equation (6)

$$\tau(p^R) = \frac{(H-L)^2 + \rho(v_l + L^2)}{\rho(v_l + L^2) + (1-\rho)(v_h + L^2)},$$

the probability of retraining at the posterior belief $p(H) = p^R$ does not depend on either cost c or probability q_L. Hence, the probability with which the firm retrains at the cutoff belief p^R is unaffected by the change of retraining cost. At the cutoff p^R the probability $\tau(p^R)$ is determined solely by the on-path indifference condition of the low-bias type.

The ex-ante probability of retraining, $Pr(r=1) = \frac{\pi}{p^R}\tau(p^R)$, is decreasing in c:

$$\frac{\partial Pr(r=1)}{\partial c} = -\frac{\pi}{(p^R)^2}\tau(p^R)\frac{\partial p^R}{\partial c} < 0.$$

Since the low-bias type chooses action H with lower probability and $\tau(p^R)$ is unchanged, the likelihood of corrective action declines as a result of the agent's change in strategy.

Next, consider a change in the agent's types distribution described by the prior belief π. Holding other parameters fixed, the cutoff belief p^R is independent of π. It enters the equation for the pooling probability q_L directly and results in:

$$\frac{\partial q_L}{\partial \pi} = \frac{1-p^R}{(1-\pi)^2 p^R} > 0.$$

As the probability of the high-bias agent increases, the low-bias type can increase the probability of imitation without making the retraining and the following expanded delegation unprofitable for the firm. Then, from the expression for $Pr(r=1)$ it can be noted that:

$$\frac{\partial Pr(r=1)}{\partial \pi} = \frac{\tau(p^R)}{p^R} > 0,$$

which indicates that the probability of retraining increases in the prior belief.

4.2.2 Firm's payoff implications

The availability of the retraining technology changes the structure of the equilibrium of the game and affects the residual bias of an agent. To isolate the firm's payoff implications of the technology, consider the corresponding equilibria with and without the retraining for the primitives under which the same equilibrium structure is sustained in both cases.

Corollary 4 (Firm payoff implications)**.** *Consider the same parameters of the model for both the retraining case and the no-retraining benchmark. If the same equilibrium structure is sustained in both cases, then:*

In a separating equilibrium, retraining increases the firm's payoff,

$$U_F^{R,S} - U_F^{NR,S} = \pi\left[R(1) - c\right] > 0.$$

In a semi-separating equilibrium, retraining decreases the firm's payoff,

$$U_F^{R,SS} - U_F^{NR,SS} = -s_l(H^2 - L^2)\left[\frac{\pi}{p^R} - \frac{\pi - p^D}{1-p^D}\right] + \left(1 - \frac{\pi}{p^R}\right)\rho s_l(v_l - L^2) < 0.$$

The signs are established in Appendix A.3.

In the case of a separating equilibrium the advantage of the retraining technology is clear. Both types choose their respective actions in period one, and in the benchmark case only the low-bias agent is delegated a low-stake task in period two. However, once the retraining is available, when the firm is certain that it faces the high-bias agent it can remove the bias and delegate all the tasks in the second period. Therefore, the corrective action allows the firm to increase its expected payoff.

In the semi-separating case, retraining reverses the direction of imitation. In the no-retraining benchmark, the high-bias type may imitate the low-bias type, while with

retraining the low-bias type may choose action H with a positive probability. In such a situation, the firm's expected first-period loss increases. Moreover, following action H, the firm has no net gain from retraining, as the gain from correction is exactly offset by its cost, $R(p^R) = c$. The only second-period gain comes from identifying a low-bias type following action L, which allows for delegation of a low-stakes task without retraining. As shown in Corollary 4, this gain is outweighed by the additional initial distortions generated by the agent's period one behavior. Hence, even though the corrective action lowers the expected mean squared error, in semi-separating comparison the availability of retraining lowers firm's payoff. Within this comparison, in the semi-separating case the firm would benefit from being able to commit not to use corrective action as it would revert the game back to the no-retraining benchmark.

Within each equilibrium structure, it is worth noting the implications of a change in retraining cost c.

Corollary 5 (Firm's payoff and cost c)**.** *Consider a change in retraining cost c. Within each equilibrium structure, the firm's equilibrium payoff satisfies the following comparative statics:*

$$\frac{\partial U_F^{R,S}}{\partial c} = -\pi < 0,$$

$$\frac{\partial U_F^{R,SS}}{\partial c} = \frac{s_\ell \pi}{(p^R)^2}\left[(H^2 - L^2) + \rho(v_\ell - L^2)\right]\frac{\partial p^R}{\partial c} > 0.$$

In a separating equilibrium, an increase in cost lowers the benefit from retraining as it becomes more expensive. Both types of agent continue to reveal their true biases, but the net gain from retraining the high-bias agent decreases in c. On the other hand, in a semi-separating equilibrium an increase in cost c has a positive effect on the firm's payoff. As the cutoff belief p^R goes up, the probability of imitation q_L declines which limits the first-period distortions and increases the informativeness of the first-period action. As the low-bias type chooses action L more frequently, the firm identifies the types better and benefits from delegating low-stakes task to the low-bias agent in period two. However, the overall firm's payoff remains below the no-retraining semi-separating equilibrium level.

5 Conclusions

The paper studies the effect of the availability of a retraining technology that corrects an existing, uncertain bias of an agent in the delegation game. It characterizes separating and semi-separating equilibrium structures under the no-retraining benchmark and when the firm has access to a costly retraining technology. The paper shows that the possibility of retraining an agent may alter the resulting equilibrium structure as it breaks the monotonicity of the agent's continuation payoff as a function of the posterior belief. In the no-retraining benchmark, a high-bias agent prefers to be perceived as a *better*, low-bias type since it may secure delegation of a low-stake task. However, once the retraining is available, the low-bias type may want to induce a higher posterior belief as it increases the firm's incentive to correct the bias, and leads to an expanded future delegation.

The non-monotonicity of the continuation value has a direct consequence for agent's equilibrium strategy and firm's payoff. In the separating equilibrium, the type is recognized following the initial action and the firm retrains the high-bias agent and profits

from a wider delegation compared to the no-retraining benchmark. The implications are starker for the firm under a semi-separating equilibrium structure. In the no-retraining benchmark, the high-bias type pools on the low-bias agent's action L with some positive probability, which lowers the initial distortions. When retraining is available, the low-bias agent is incentivized to overstate its bias and choose action H with some positive probability. The low-bias agent is willing to incur a cost of imitation in order to look *worse* on-path. This results in a greater initial total distortion, while in equilibrium the firm is indifferent between eventually retraining and using its in-house process. The firm's participation comes only from correctly recognizing a low-bias agent following action L, however, it never compensates for the additional distortions compared to the benchmark. An increase in the cost of retraining may partially mitigate the loss as it decreases the probability of a bias exaggeration by the low-bias type. Within the semi-separating equilibrium comparison, the firm would benefit from a commitment to never retrain as it would restore the monotonicity of the agent's continuation value.

The present setup assumes that the agent has no intrinsic value in preserving its initial bias. The agent's objective is defined relative to its contemporaneous bias in each period. An alternative specification could introduce a cost to the agent of having its objective function altered. For instance, retraining could impose a direct cost on the agent that increases with the magnitude of the correction. Such a modification would result in a trade-off between the benefits of future delegation and the cost of changing the objective. As a result, both types could have an incentive to avoid corrective action, even if that reduces future delegation. This could alter the incentive structure and the equilibrium strategies characterized in the present model. The alternative specification could provide some further insights regarding the AI agent's strategic response to the possibility of having its goal function modified.

An interesting avenue to explore further is the impact of the retraining mechanism presented in an infinite-horizon setting. If a single retraining decision permanently improves the agent's expected alignment, the gain from corrective action and future delegation could be substantially greater than in the baseline model. The current two-period setup can also be interpreted as a single cycle within an infinite horizon problem with a recurring reset of the agent's bias, for instance, due to AI system updates or the introduction of new versions. If each update resets the bias and restores uncertainty about the agent's type, the firm repeatedly faces the delegation problem, and the mechanism identified in the present analysis may recur following every reset. A more natural extension would allow the agent's bias to reset with some probability in each period rather than in fixed cycles. In such a setting, the availability of retraining could still result in bias exaggeration by the low-bias type. However, whether such behavior is sustained in equilibrium requires further investigation, as the uncertain reset would affect continuation values, and in consequence, the strategic incentives of the agent and the firm.

The paper provides insights into the behavior of an agent in a delegation setting when its bias may be altered by a retraining technology. The implications of such technology go beyond simply enhancing the future agent's performance. It reshapes the incentives of both types, and in certain cases may generate *undesirable* strategic behavior. The results point to the theoretical possibility of an agent strategically creating a stronger distortion to convince the firm to apply costly corrective action, which becomes imperfect due to the uncertainty. The mechanism presented here connects the analysis to the growing literature on the strategic and deceptive behavior of AI agents.

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A Proofs

A.1 Proof of Lemma 3

Proof. First, from the condition in (4) there is no $p \in [0, \pi]$ for which $R(p) \geq c$. Therefore, the cutoff satisfies $p^R > \pi$.

Suppose that $\pi < \frac{1}{2}$. By the second part of Assumption 1, $p^D < \pi$. For $p \in [\pi, \frac{1}{2}]$ there is no delegation without the retraining. Hence, for every p in that interval $R(p)$ is decreasing as $B_R(p)$ increases. Therefore, $R(\frac{1}{2}) < R(\pi) < c$. From the condition in (4), the cutoff $p^R > \max\{\frac{1}{2}, \pi\}$. If $\pi \geq \frac{1}{2}$, then it follows immediately that $p^R > \frac{1}{2}$.

Furthermore, the second inequality in the condition in (4), $c < R(1)$, immediately implies that $p^R < 1$.

Then, it is necessary to show that the cutoff exists and is unique on the interval. As $\max_{p\in[0,\pi]} R(p) < c$ and $c < R(1)$, and $R(p)$ is continuous and strictly increasing for $p > \frac{1}{2}$, by the intermediate value theorem there exists a unique p^R for which $R(p^R) = c$.

As $R(p)$ is continuous and strictly increasing for $p > \frac{1}{2}$, $p^R = R^{-1}(c)$ is increasing in c, and by continuity $p^R \to 1$. □

A.2 Non-emptiness of benchmark and retraining equilibrium regions

Proof. First, consider the equilibrium structures shown in Proposition 1.

- Separating equilibrium: There are two conditions apart from Assumption 1 that need to be satisfied at the same time:

  1. $(H - L)^2 \geq \rho\left(v_\ell + H^2\right)$; and

  2. $\pi \leq \frac{p^D(1+\rho)}{1+\rho p^D}$.

  First, select parameters H, L, v_l and v_h to satisfy the first part of Assumption 1. Then, consider ρ sufficiently close to 0, such that $\rho > 0$ and

  $$(H - L)^2 > \rho\left(v_\ell + H^2\right),$$

  which ensures condition (1) holds. Then, for any $\rho > 0$ the following is true:

  $$\frac{p^D(1+\rho)}{1+\rho p^D} > p^D.$$

  Therefore, there is some π such that:

  $$p^D < \pi < \frac{p^D(1+\rho)}{1+\rho p^D},$$

  where the first inequality satisfies part two of Assumption 1, and the second inequality satisfies firm's participation constraint.

- Semi-separating equilibrium: There are two conditions apart from Assumption 1 that need to be satisfied:

1. $(H-L)^2 < \rho\left(v_\ell + H^2\right)$; and
2. $\pi \leq p^D(2-p^D)$.

First, select parameters H, L, v_l and v_h to satisfy the first part of Assumption 1. Then, observe that the following always holds:

$$(H-L)^2 < v_l + H^2.$$

Then, pick ρ sufficiently close to 1 such that:

$$(H-L)^2 < \rho(v_l + H^2),$$

which satisfies condition 1. Moreover,

$$p^D(2-p^D) > p^D,$$

hence there exists π such that:

$$p^D < \pi < p^D(2-p^D),$$

where the first inequality satisfies part two of Assumption 1, and the second inequality satisfies firm's participation constraint.

Both equilibrium structures discussed in Proposition 1 exist for some non-empty set of parameters.

Then, consider the equilibrium structures shown in Proposition 2. This part requires two additional conditions described in equations (3) and (4).

Start by considering condition in (4). From the proof of Lemma 3:

$$R(1) > R(p), \quad \forall p < 1.$$

Therefore, there exists some cost c such that:

$$R(p) < c < R(1), \quad \forall p \in [0, \pi],$$

and the condition in (4) is satisfied. By Lemma 3 such cost induces a unique cutoff $p^R \in (\max\{\pi, \frac{1}{2}\}, 1)$, and as $c \to R(1) \implies p^R \to 1$.

Then, consider condition in (3):

$$\frac{(H-L)^2}{4} < v_h,$$

and select parameters L, H such that $(H-L)$ goes towards 0, and v_l and v_h such that the above is satisfied together with part one of Assumption 1:

$$\frac{(H-L)^2}{4} < v_h < L^2 < v_l < H^2.$$

- Separating equilibrium: Apart from the above conditions and the second part of Assumption 1, the separating equilibrium needs to satisfy:
  1. IC-L: $(H-L)^2 \geq (1-\rho)(v_h + L^2)$; and
  2. PC-F: $s_l\left[B_0(\pi) - v_l\right] \leq (1-\pi)\rho s_l\left(v_l - L^2\right) + \pi\left[R(1) - c\right].$

Then, for ρ sufficiently close to 1, IC-L condition holds:

$$(H-L)^2 > (1-\rho)(v_h + L^2).$$

Finally, choose π sufficiently close to p^D such that $\pi > p^D$. This satisfies part two of Assumption 1. Moreover, for such π, $B_0(\pi) - v_l \to 0$, hence the LHS of firm's participation condition goes to 0 and:

$$s_l \left[B_0(\pi) - v_l\right] < (1-\pi)\rho s_l \left(v_l - L^2\right) + \pi \left[R(1) - c\right],$$

as the continuation gain remains strictly positive.

- Semi-separating equilibrium: Apart from the above conditions and the second part of Assumption 1, the semi-separating equilibrium needs to satisfy:

  1. IC-L: $(H-L)^2 < (1-\rho)(v_h + L^2)$; and
  2. PC-F: $L^2 + \frac{\pi}{p^R}(H^2 - L^2) - v_l \leq \frac{p^R - \pi}{p^R}\rho\left(v_l - L^2\right)$.

  Use the same parameters to satisfy part one of Assumption 1 and condition in (3) as described above. Then, as $(H-L)^2$ is close to 0, v_h can be picked to satisfy:

$$(H-L)^2 < v_h + L^2,$$

and for sufficiently small ρ, the IC-L condition is satisfied:

$$(H-L)^2 < (1-\rho)(v_h + L^2).$$

Then, the PC-F condition can be rewritten as follows. Using the definition $p^D = \frac{v_l - L^2}{H^2 - L^2} \Leftrightarrow v_l - L^2 = p^D(H^2 - L^2)$, the LHS is equivalent to:

$$(H^2 - L^2)(\frac{\pi}{p^R} - p^D).$$

Using the same definition of p^D, the RHS gives $\frac{p^R - \pi}{p^R}\rho\left(v_l - L^2\right) = (1 - \frac{\pi}{p^R})\rho p^D(H^2 - L^2)$. Dividing both LHS and RHS by $(H^2 - L^2)$ yields:

$$(\frac{\pi}{p^R} - p^D) \leq (1 - \frac{\pi}{p^R})\rho p^D.$$

Then, this can be rewritten as:

$$\pi \leq p^R \frac{p^D(1+\rho)}{1+\rho p^D},$$

where for costs c, such that $c \to R(1)$, the cutoff p^R approaches 1. As:

$$\frac{p^D(1+\rho)}{1+\rho p^D} > p^D,$$

for any $\rho > 0$, choose π sufficiently close to p^D such that $\pi > p^D$. Then, as p^R approaches 1, $p^R \to 1$, the participation constraint and part two of Assumption 1 are satisfied:

$$p^D < \pi < \frac{p^D(1+\rho)}{1+\rho p^D}.$$

Therefore, both parameter regions in Proposition 2 are non-empty. □

A.3 Proof of Corollary 4

Proof. In the case of the separating equilibrium, the increase in the firm's payoff follows immediately from the assumption about the cost $c < R(1)$:

$$U_F^{R,S} - U_F^{NR,S} = \pi[R(1) - c] > 0.$$

Then, in the case of the semi-separating payoff comparison, first, using the definition of p^D established in the no-retraining benchmark:

$$v_l - L^2 = p^D(H^2 - L^2).$$

Hence, the payoff difference in Corollary 4 can be rewritten as follows:

$$\begin{aligned} U_F^{R,SS} - U_F^{NR,SS} &= -s_l(H^2 - L^2)\left(\frac{\pi}{p^R} - \frac{\pi - p^D}{1 - p^D}\right) + \left(1 - \frac{\pi}{p^R}\right)\rho s_l(v_l - L^2) \\ &= -s_l(H^2 - L^2)\left(\frac{\pi}{p^R} - \frac{\pi - p^D}{1 - p^D}\right) + \left(1 - \frac{\pi}{p^R}\right)\rho s_l p^D(H^2 - L^2) \\ &= s_l(H^2 - L^2)\left[-\left(\frac{\pi}{p^R} - \frac{\pi - p^D}{1 - p^D}\right) + \rho p^D\left(1 - \frac{\pi}{p^R}\right)\right]. \end{aligned}$$

Then, expanding and rewriting the above yields:

$$U_F^{R,SS} - U_F^{NR,SS} = -s_l(H^2 - L^2)\left[\frac{\pi(1 - p^R)}{p^R(1 - p^D)} + \frac{p^D(p^R - \pi)}{p^R}\left(\frac{1}{1 - p^D} - \rho\right)\right] < 0,$$

where the inequality follows from $p^R < 1$, $p^R > \pi$, $p^D \in (0, 1)$, and $\rho < 1$. □

A.4 Proof of Corollary 5

Proof. Using the expressions derived in Corollary 4, differentiate both with respect to the cost of retraining c.

In a separating case:

$$\frac{\partial U_F^{R,S} - U_F^{NR,S}}{\partial c} = \frac{\partial \pi[R(1) - c]}{\partial c} = -\pi < 0.$$

Since $U_F^{NR,S}$ is independent of cost, the above can be rewritten as:

$$\frac{\partial U_F^{R,S}}{\partial c} = -\pi < 0.$$

In a semi-separating case, the payoff loss under retraining is described by:

$$U_F^{R,SS} - U_F^{NR,SS} = -s_l(H^2 - L^2)\left[\frac{\pi}{p^R} - \frac{\pi - p^D}{1 - p^D}\right] + \left(1 - \frac{\pi}{p^R}\right)\rho s_l(v_l - L^2).$$

As the change in cost affects the above expression only via p^R, the derivative can be written as:

$$\frac{\partial U_F^{R,SS}}{\partial c} = \frac{\pi s_l}{(p^R)^2}[(H^2 - L^2) + \rho(v_l - L^2)]\frac{\partial p^R}{\partial c} > 0,$$

where the inequality follows from Assumption 1 and Lemma 3.

□